

Solution Requirements

Date	29/09/2026
Team ID	UNM1425COMICCRAFT
Project Name	COMICCRAFT
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No login is required in the current version. ComicCraft is a single-user web app opened at the home page (index.html). The Gemini API key is not entered by users; it is kept on the server (environment variable) and never shown in the browser.	Low
2	Authorization levels	One role only: User , who can submit a story, preview it and download the PDF. The /test-image route is a developer utility for testing image generation and is not part of the user flow.	Low
3	External interfaces	Google Gemini API: Gemini Flash (models/gemini-1.5-flash) for the 5-panel outline and Gemini Pro (models/gemini-1.5-pro) for narration and dialogue, via google-generativeai. Stable Diffusion (runwayml/stable-diffusion-v1-5) through Hugging Face Diffusers for panel images. FPDF library for PDF export.	High

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
4	Transactions processing	No payments. Each request is one generation transaction: the form (POST /generate) or JSON (POST /generate-comic/json) input goes through generate_outline() → generate_story() → generate_image() for every panel → build_comic_layout() → save_pdf(). Failures are handled with HTTPException and try-except.	High
5	Reporting	Comic preview page (comic_preview.html) with all panels, images and text. PDF export of the full comic with a timestamped filename saved in static/exports. Export success page (export_success.html). JSON response with layout data and PDF path for API clients.	High
6	Business rules, etc.	The user provides a story prompt, character name, setting, tone and art style. Every comic has exactly 5 panels; each panel has a number, title, scene description and image prompt. Changing the tone or art style regenerates the whole pipeline (outline, story and images). Image filenames are sanitized before saving.	High
7	Compliance to laws or regulations	No personal data is collected or stored, so GDPR-type obligations are minimal. The system follows the Google Gemini API terms of use and the Stable Diffusion v1-5 model license (CreativeML OpenRAIL-M), and must not be used to create harmful or illegal content.	Medium
8	Other	Input validation with the Pydantic PromptRequest schema and Form(...) parameters; friendly error messages when an AI call fails; Jinja2 templates for a consistent web interface; local deployment with Uvicorn.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Text generation (outline and story) should return quickly; image generation is the slowest step because Stable Diffusion creates one image per panel. Pages themselves must load fast.	Pages load in < 2 s; outline + story in < 30 s; each panel image in < 60 s on a GPU; PDF export in < 10 s

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
2	Scalability	Runs as a local deployment on Uvicorn (ASGI). It can handle several users by adding workers or a GPU machine, and generated files are kept in separate folders (static/panels, static/exports).	<i>Supports about 5 concurrent users on one machine; more with extra workers</i>
3	Security & Data Privacy	API keys stay in environment variables, not in code or templates. User input is validated and filenames are sanitized. No personal data is stored beyond the generated comic files.	<i>0 API keys in source code; only validated inputs accepted; no personal data saved</i>
4	Reliability & Availability	The app stays available while the server runs, and returns a clear error message if the Gemini API or the image model fails, instead of crashing.	<i>95% of valid requests produce a complete 5-panel comic; no unhandled crashes</i>
5	Usability & Accessibility	A simple HTML form with five labelled fields, a readable comic preview and a one-click PDF download, so non-artists can use it without training.	<i>A new user creates a comic in < 3 steps; readable colour contrast and labelled form fields (aiming for WCAG 2.1 AA)</i>
6	Other	Maintainability: modular code (gemini_flash.py, gemini_pro.py, image_generator.py, layout_builder.py, exporters.py, routes.py). Portability: runs on Windows or Linux with Python and a virtual environment.	<i>Each feature in its own module; setup completed with one pip install command</i>